

Enhancing Temporal Classification of AAR Parameters in EEG single-trial analysis for Brain-Computer Interfacing

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Abstract— Adaptive autoregressive (AAR) coefficients provide dynamic spectral information in EEG single-trial analysis. In this paper we propose a temporal evidence accumulation framework to enhance classification of AAR features. The results for a single subject, using 280 trials, indicate distinct improvements over a conventional method of temporal classification. We illustrate how the framework is applicable to AAR features, as well as to wavelet features as reported in [13]. These findings put the two time-frequency features on equal footing for comparison in this context.

IndexTerms—Brain-computer interface (BCI), electroencephalogram (EEG), adaptive autoregressive model (AAR), motor imagery.

I. INTRODUCTION

BRAIN-COMPUTER interfaces (BCI) allow direct communication with the brain by extracting information from its electrical activity. This technology has significant implications for the physically disabled, particularly those who cannot produce any motor output whatsoever. In this paper we investigate classification of imagined left vs. right hand movements, commonly referred to as motor imagery, using EEG originating from the sensorimotor cortices. This thought-task is particularly appealing for BCI's that carry out motor related activity, as the required thoughts are closely related to the required action.

Motor imagery is accompanied by a contra-lateral attenuation in the alpha and lower beta bands known as an event-related desynchronization (ERD) [1]. When cortical areas are less active an increase in amplitude is observed known as an event-related synchronizations (ERS) [2]. We may exploit these characteristics to distinguish left and right hand movements by using time-frequency analysis.

Autoregressive parameters (AR) have been used as features with success in EEG analysis [3],[4] and the AR model has several advantages: it is a maximal entropy spectral estimator and only a few parameters are required to describe spectral information [5]; it does not require *a priori* knowledge of relevant frequencies, as they can vary from subject-to-subject; it describes the stochastic nature of EEG quite well [6]; and the methods for their computation are

well studied. Conventional approaches to address non-stationary behavior of EEG have been to apply a moving window of AR estimates such as in [3] and [4]. Depending on the size and shift of the window, traditional AR estimation algorithms can become computationally intensive. Instead, we may estimate non-stationary AR parameters by inferring a dynamic model and using Kalman filtering [7]. Such an approach is referred to as adaptive autoregressive modeling (AAR) and has been successfully used in left-right single-trial motor imagery classification [8], [9].

In this paper we show how AAR parameters can be used more effectively in single-trial classification by combining decisions made over the duration of the trial. This framework uses evidence accumulated over time to make a decision about a trial. Although data was analyzed off-line in this study, all signal processing is causal and applicable on-line. We consider a single subject imagining left and right hand movements using similar protocols as in [8] and [9]. The dataset was provided by the Department of Medical Informatics, Institute for Biomedical Engineering, University of Technology, Graz, Austria. The EEG was sampled at 128 Hz from three channels (C3, Cz, and C4) that were band-pass filtered allowing frequencies between 0.5 to 30 Hz. The dataset consists of 280 trials that are 9s long each. The first 3s is an idle preparation period, at which point a visual cue in the form of an arrow appears pointing either to the left or the right. The user was instructed to perform the imagination task according to the direction of the arrow for the next six seconds. During this time the system provided the user with visual feedback using an on-line classifier.

Although in this paper we focus on AAR features specifically, it is within the context of determining which time-frequency feature extraction approach is most appropriate for motor imagery EEG in BCI, and is meant to contribute to that debate. It is also important to realize that the classification we do in this synchronous protocol can be applicable to asynchronous protocols after the successful detection of a motor event.

TABLE I
SUMMARY OF AAR KALMAN FILTER PARAMETERS

Symbol	Definition	Dimensions
$W(n)$	Process noise variance	scalar
$R(n)$	Variance of measurement noise	scalar
$\mathbf{z}(n)$	Estimate of AAR parameters (state)	$p \times 1$
$\mathbf{Y}(n)$	Measurement matrix (last p samples of y)	$p \times 1$
$\mathbf{G}(n)$	Kalman gain	$p \times 1$
$v(n)$	Innovation process	scalar
$Q(n)$	Innovation process variance	scalar
$\mathbf{P}(n)$	Predicted state-error correlation matrix	$p \times p$
C	Update coefficient	scalar

II. METHOD

A. AAR estimation using Kalman Filtering

A p^{th} order AAR model can describe a single EEG sensor using

$$y(n) = \mathbf{z}(n)^T * \mathbf{Y}(n-1) + v_m(n) \quad (1)$$

where n is the discrete time index; y is the EEG sample; $\mathbf{z}$ is a p -element column vector of the autoregressive coefficients; $\mathbf{Y}$ is a column vector of the last p EEG samples; and v_m is a zero-mean, white-noise process [8]. Notice that the autoregressive coefficients change with time to capture the dynamics of the EEG. In the context of the Kalman Filter [10], (1) is the measurement model and the following is the process or dynamics model:

$$\mathbf{z}(n) = \mathbf{z}(n-1) + v_p(n). \quad (2)$$

where v_p is zero-mean white noise. Notice that we model the system as a random walk and assume small changes in the state.

The Kalman filter algorithm is given below in update-update form; there is little advantage in presenting the algorithm in both prediction and update steps, as it is commonly written, since our model does not have a process transition matrix (See table 1 for description of parameters):

$$Q(n) = (1 - uc) * Q(n-1) \quad (3)$$

$$\mathbf{G}(n) = \mathbf{P}(n-1) \cdot \mathbf{Y}(n) / Q(n) \quad (4)$$

$$\hat{\mathbf{z}}(n+1) = \hat{\mathbf{z}}(n) + \mathbf{G}(n)v(n) \quad (5)$$

$$W(n) = C * \text{trace}(\mathbf{P}(n-1)) / p \quad (6)$$

$$\mathbf{P}(n) = \mathbf{P}(n-1) - \mathbf{G}(n)^T \cdot \mathbf{Y}(n-1) \cdot \mathbf{P}(n-1) + W(n) \quad (7)$$

The update coefficient C and model order p are to be selected so that the AAR estimates best describe the EEG signal y . In [11] they propose the relative error variance (REV) as a minimization criterion for selecting these parameters and it is given by

$$REV = \frac{\sum v(n)^2}{N} / \text{var}(y), \quad (8)$$

which is the mean squared error normalized by the signal

variance. N is the total number of samples in the trial. In this study we use the REV criterion as a guideline for selecting p and C but do not necessarily use the parameter values that minimize REV. Lower REV values do not always justify larger model orders, if the increase in complexity and computation do not translate to significant improvement at the classification stage. For the subject in this study we estimate AAR features from C3 and C4 and use a model order $p=3$ and update coefficient $C = 0.0078125$, selected based on the REV criterion.

B. Classification with Time Accumulated Evidence

The objective is to classify the trial with a significant degree of certitude as quickly as possible. To address the non-stationary EEG the classifier is trained separately at each discrete instant in time. Typically such classification techniques have used linear-discriminant analysis (LDA) trained and tested at each instant in time independently from one another. In this paper we propose that this approach does not take advantage of the accumulated evidence the AAR features are providing over time. We apply a temporal evidence accumulation framework based on one proposed in [13], and investigate its success with AAR features.

Let $\mathbf{z}_{t,3}(n)$ and $\mathbf{z}_{t,4}(n)$ denote row vectors of the estimated AAR parameters for trial t at time n for electrodes C3 and C4, respectively. We concatenate the two vectors to form a single 6-dimensional feature vector giving

$$\mathbf{z}_t(n) = [\mathbf{z}_{t,3}(n), \mathbf{z}_{t,4}(n)]^T. \quad (9)$$

Labeled training data from 10-fold cross validation of the dataset is used to infer a class conditional 6-D Gaussian distribution for both classes Left (L) and Right (R). Thus we have the probability density

$$p(\mathbf{z}(n) | c) = \frac{|\Sigma_{n,c}|^{-1/2}}{4\pi^2} \exp\left\{-\frac{1}{2}(\mathbf{z}(n) - \mu_{n,c})^T \Sigma_{n,c}^{-1}(\mathbf{z}(n) - \mu_{n,c})\right\} \quad (10)$$

where $c \in \{L, R\}$. The mean $\mu_{n,c}$ and covariance $\Sigma_{n,c}$ are estimated from the training data for c . We may classify $\mathbf{z}_t(n)$ by calculating the probability of belonging to a class using

(10) and Bayes theorem as

$$p(c | \mathbf{z}(n)) = \frac{p(\mathbf{z}(n) | c)}{p(\mathbf{z}(n) | L) + p(\mathbf{z}(n) | R)}. \quad (11)$$

In order to combine information throughout time the decision made at the current time $n = n_t$ is a weighted average of all the decisions for $n \leq n_t$; thus we have

$$p(c | \mathbf{z}(1) \dots \mathbf{z}(n_t)) = \frac{\sum_{n \leq n_t} w_t p(c | \mathbf{z}(n))}{\sum_{n \leq n_t} w_t} \quad (12)$$

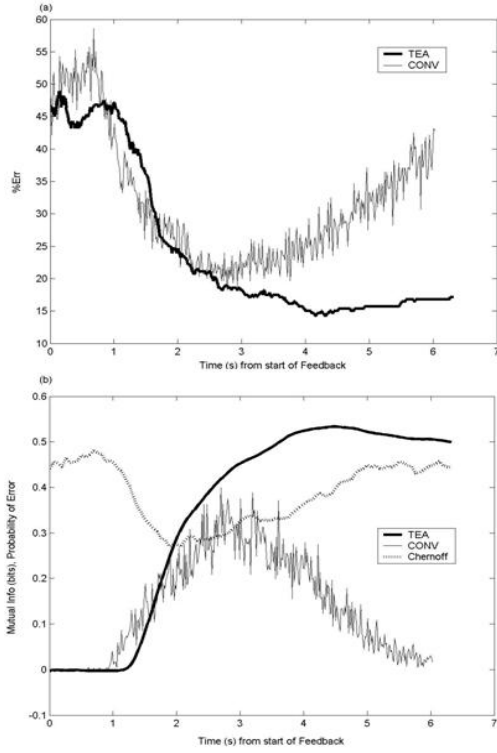


Fig. 1. (a): Time course of percent error from 10-fold cross validation for the TEA and CONV approaches to classification of AAR features. (b): The time course of the Chernoff bound (dotted line) and the mutual information between the decision output and classes for the TEA (thick line) and CONV (thin line) methods.

where w_t reflects our certitude in the decision making ability of the classifier at time n_t , and is derived from the probability of misclassification [13]. Although we cannot find the probability of misclassification directly we may calculate its upper limit using the Chernoff bound [14]; an advantage of using an explicit probabilistic approach to classification.

III. RESULTS AND DISCUSSION

A. Performance Measures

We perform two experiments on the data set: the temporal evidence accumulation (TEA) approach and the conventional (CONV) approach. CONV involves LDA classification of the AAR parameters at each discrete-time instant (see [8],[9] and [12] for more details). We employ three measures of performance using 10-fold cross validation: the time course of the percent misclassified (%Err); the signed decision magnitude, $D(n)$; and the mutual information, $I(n)$ [12].

$D(n)$ reflects both the classifier's decision by its sign ($D(n) > 0 \rightarrow R$, $D(n) < 0 \rightarrow L$) and certitude by its magnitude [8]. For CONV, the signed decision magnitude $D_{\text{CONV}}(n)$ is the shortest distance between $\mathbf{z}(n)$ and the LDA boundary and, for TEA, $D_{\text{TEA}}(n)$ is defined as [13]

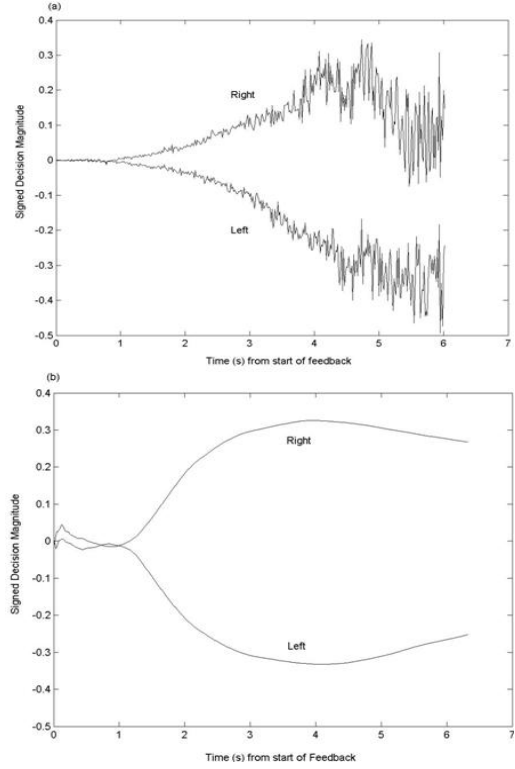


Fig. 2. (a): Time course of left and right decision outputs for the CONV (a) and TEA (b) methods. For CONV the distance to classification boundary has been scaled to be numerically comparable to TEA.

$$D_{\text{TEA}}(n) = 1 - 2 \cdot p(L | \mathbf{z}(1) \dots \mathbf{z}(n_c)) . \quad (13)$$

The reasons for examining $D(n)$ are two-fold: it gives a good indication of the separability of classes and the classifier's confidence in its decision; second, it collapses the problem into a single dimension lending itself well to computational simplicity and efficiency for information theoretic measures. $D(n)$ can be modeled as having two processes: one being the motor imagery task and the other being noise [12]. Information theory may be used to evaluate how much information $D(n)$ provides about the motor imagery task. The entropy of $D(n)$ is a measure of its uncertainty at time n and, inferring a Gaussian distribution, is defined as

$$H(D(n)) = \frac{1}{2} \log(2\pi e \sigma^2), \quad (14)$$

where σ^2 is the variance of $D(n)$. The reduction in entropy from $H(D(n))$ to the within-class entropy $H(D(n)|c)$ is the mutual information [12]. It is a measure of how closely related $D(n)$ is to the motor imagery classes, and thus, quantifies the amount of information in $D(n)$ that is relevant to the problem. The mutual information can be calculated by

$$I(D(n)) = \frac{1}{2} \log \left[\frac{\sigma^2}{0.5(\sigma_L^2 + \sigma_R^2)} \right], \quad (15)$$

where σ_L^2 and σ_R^2 are the class conditional variances of $D(n)$, assuming the noise process for $D(n)$ is also Gaussian. If this assumption is incorrect, $I(D(n))$ in (15) is in fact the upper limit on the mutual information [12].

B. Results

The time course of the classification results and mutual information for both experiments are depicted in figure 1. The TEA method reaches a minimum error of 14.3% while CONV achieved a very inconsistent minimum of 18.2%. If one were to consider a moving average of the CONV error curve the minimum would be 21.6%. More significant than the reduction in minimum error is the substantial difference in variability throughout the time course. This is an important improvement in the TEA approach for enhanced predictability of the BCI system.

The advantages of the TEA approach are more compelling in the plot of mutual information. Notice that $I_{TEA}(n)$ reaches a maximum of 0.53 compared to 0.41 for $I_{CONV}(n)$. The latter maximum is inconsistent as there is more variability in $I_{CONV}(n)$. If we again consider a moving average of the curve, $I_{CONV}(n)$ has a maximum of only 0.33. In addition to attaining higher maximal mutual information, I_{TEA} demonstrates a more rapid rate of increase. This is a useful improvement in BCI applications where timely response to user input is desirable.

The time course of $D_{TEA}(n)$ and $D_{CONV}(n)$ are depicted in figure 2. Notice the greater consistency in $D_{TEA}(n)$ and faster rate at which the classification problem is more separable.

C. Implications in a BCI Context

The results for both TEA and CONV reflect an improvement in the separability of the classes over time followed by degradation. For CONV this is evident both in %Err and $I_{CONV}(n)$ in Figure 1 (a) and (b), respectively. For TEA this is evident in the Chernoff bound. This may be a result of a decrease in concentration and focus on the part of the user. There is an important difference in the way the two algorithms handle this problem. The CONV approach does not make use of the knowledge the training data provides about this degradation. The TEA method, however, measures uncertainty in the training data enabling it to predict the degradation in features in the latter part of the trial.

The results indicate that classification of AAR features in single-trial analysis benefit considerably from the temporal evidence accumulation framework. Although the framework was proposed for wavelet features in [13], we have shown how it can enhance classification of AAR features. This is an important finding in single-trial time-frequency analysis as it puts AAR features and wavelet features on equal footing for comparison in this context. If, using this framework, performance of AAR features is comparable to wavelet features, AAR modeling is an attractive alternative since it

does not require prior knowledge of subject-specific relevant frequency bands. Not only can the AAR approach be more versatile between subjects, but also within the same subject for whom relevant frequency bands may change over the short or long-term.

IV. CONCLUSION AND FUTURE WORK

The best method for time-frequency analysis in EEG single-trial analysis continues to be debatable. Is it wavelet, AAR or some other time-frequency paradigm? We have shown, for the subject studied, that classification of AAR features can be improved considerably using the TEA approach. Future work will compare extensively performance of the TEA method as applied to AAR and wavelet for several subjects. We will not only compare them but also investigate complimentary time-frequency information between the two feature sets.

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